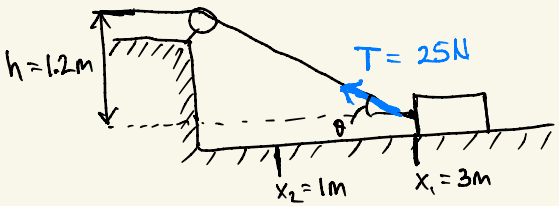
Ch7: Pr 42



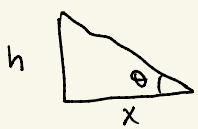
what is ΔKE?

$dW = T_x\,dx$

$dW = T\cos\theta\,dx$

} only x-comp of T does work

3gecb $\quad \cos\theta = \dfrac{x}{\sqrt{h^2+x^2}}$



$\cos\theta = \dfrac{x}{\sqrt{x^2+h^2}}$, $\quad dW = T\dfrac{x}{\sqrt{x^2+h^2}}\,dx$

$$W = \int dW = \int_{x_1}^{x_2} T\,\frac{x\,dx}{\sqrt{x^2+h^2}}$$

Use Method of Substitution: let $x^2+h^2 = u$

$\dfrac{du}{dx} = 2x \rightarrow x\,dx = \dfrac{du}{2}$.

$$W = \int_{x_1}^{x_2} T\,\frac{x\,dx}{\sqrt{x^2+h^2}} = \int_{u_1}^{u_2} T\,\frac{du/2}{\sqrt{u}} = T\,\cancel{2}\sqrt{u}\,\cancel{\tfrac{1}{2}}\Big|_{u_1}^{u_2}$$

Note: $\displaystyle\int \frac{du}{\sqrt{u}} = \int u^{-1/2}\,du = \frac{u^{1/2}}{1/2} + C = 2\sqrt{u} + C.$

$$W = T\sqrt{x^2+h^2}\Big|_{x_1}^{x_2}$$

$$= T\left(\sqrt{x_2^2+h^2} - \sqrt{x_1^2+h^2}\right)$$

$$= (25N)\left[\sqrt{1m^2+(1.2m)^2} - \sqrt{(3m)^2+(1.2m)^2}\right]$$ #

we know $W > 0$, so take |abs value| $\left|\sqrt{x^2+h^2}\right|_{x_1}^{x_2}$